Replenishing perishables on shared delivery routes: Stock-age-based inventory policies, stochastic lookahead and decision-focused learning

Abstract

Fresh products age on the shelf, and deliveries to stores share a capacitated fleet. We study how contextual demand forecasts should be turned into replenishment decisions in a periodic-review perishable inventory-routing system with FIFO issuing, lost sales and shelf capacity. Nine policies are compared in closed loop with identical information: base-stock rules with and without estimated waste, the deliver-up-to-level rule of Crama et al. (2018), rolling-horizon inventory-routing models driven by mean, quantile, robust and scenario forecasts (Scenario Predict-then-Optimize, Jia et al., 2025), a decision-focused learning (DFL) policy that learns context- and stock-age-dependent planning targets, and a perfect-information reference. All lookahead decisions are solved to a proven 0.5% gap. We show that the inventory recourse admits FIFO-optimal solutions once early disposal is allowed, bound the marginal value of a delivery, and show that DFL stationary points reduce to the perishable newsvendor fractile. Across ten settings of a ten-store network with synthetic contextual demand, simple stock-age-based rules with daily delivery are cheapest in nine, 2.4–8.9% below the best integrated policy; only with expensive routing does scenario-based lookahead win. No integrated policy dominates: DFL is best in five settings and scenario lookahead in four. DFL improves the deterministic quantile lookahead in nine settings (significantly in five) by learning that only the implemented day needs a high safety quantile. Integrated plans also concentrate stock-outs at peripheral stores.

Keywords: Perishable inventory, Inventory routing, Waste, Lost sales, Base-stock policies, Decision-focused learning

1. Introduction

Fresh food is replenished daily, its stock ages every night, and whatever is not sold within its shelf life is thrown away. Retailers and their distributors therefore face the classical perishable inventory trade-off between waste and lost sales (Nahmias, 1982; Bakker et al., 2012), but with two complications that single-store inventory models do not capture. First, deliveries to different stores share a small fleet: a store that is not on today's route cannot be replenished today, and a vehicle that is full cannot top up the next store. Second, retailers now

have rich contextual information—promotion calendars, weather forecasts, recent sales—that machine-learning forecasters turn into store-day demand predictions. The operational question is how such forecasts should be translated into replenishment and delivery decisions when stock ages, shelf space is limited and vehicles are shared.

Two literatures answer different halves of this question. Perishable inventory control offers simple, robust ordering rules—base-stock policies (Bu et al., 2023), and rules that anticipate the waste of the current stock (Broekmeulen and van Donselaar, 2009; Haijema and Minner, 2019)—but treats each store in isolation. Inventory routing integrates replenishment and routing, with exact methods for deterministic perishable demand (Coelho and Laporte, 2014; Alvarez et al., 2020), decomposition and order-up-to rules for stochastic perishable demand (Crama et al., 2018), robust formulations (Solyalı et al., 2012; He et al., 2024) and, recently, scenario-based predict-then-optimise policies that use machine-learning quantile forecasts (Jia et al., 2025). In parallel, decision-focused learning (DFL) trains forecasters on the cost of the decisions they induce rather than on forecast accuracy (Elmachtoub and Grigas, 2022; Mandi et al., 2024). Whether DFL adds anything once the strong inventory rules and stochastic lookahead methods are on the table has not been tested for perishable inventory routing.

This paper studies a periodic-review perishable inventory-routing system with contextual demand, lost sales, FIFO issuing and shelf capacity, and compares nine replenishment policies in closed loop, every day observing the age-structured stock and the same forecasts. The comparison includes the inventory-control rules a perishable-inventory manager would use (base-stock with and without estimated waste), the deliver-up-to-level rule of Crama et al. (2018), and rolling-horizon inventory-routing models driven by mean, quantile, robust and scenario forecasts, the last being the Scenario Predict-then-Optimize method of Jia et al. (2025). We propose a decision-focused policy, LA-DFL, that keeps a deterministic rolling-horizon plan but learns how far each planning target should deviate from the forecast, depending on context and on the age of the store's stock. The learning signal is the realised FIFO cost of the deliveries the plan induces, differentiated through a coverage layer while an exact MILP oracle fixes the routes.

The answer is more nuanced than either literature suggests. Across ten settings, daily base-stock rules that use the age of the stock, with their orders routed optimally, were the cheapest policies in nine; only when travel was expensive was the scenario-based lookahead best. Decision-focused learning improved the deterministic lookahead in nine settings and made it the best integrated policy in five, largely by learning that only the executed day needs a high safety quantile, but it did not overturn the advantage of the inventory rules.

The paper makes three contributions.

1. **An inventory-centred benchmark of replenishment policies under routing cou-**

pling. We evaluate stock-age-based inventory rules, a published stochastic PIRP rule, and integrated robust, scenario-based and decision-focused lookahead policies under one closed-loop simulator, with every lookahead solved by an exact MILP to a proven gap. Outcomes are reported in inventory terms—cost composition, fill rate, waste rate, delivery frequency and store-level service—across shelf life, cost ratio, fleet tightness, demand variability and issuing policy.

2. **A decision-focused policy grounded in perishable inventory structure.** We show that the inventory recourse admits FIFO-optimal solutions once early disposal is allowed, bound the marginal value of a delivery by the lost-sales and waste penalties, and prove that stationary points of the decision-focused objective satisfy the stochastic optimality conditions of the fixed-route problem, reducing to the perishable newsvendor fractile for one-day shelf life. These results also correct claims of an earlier version of this work, and all of them are verified numerically.

3. **Managerial evidence on when integration, stochastic lookahead and learning pay off.** The results show when simple stock-age-based rules suffice and when integration with routing pays, why deterministic plans over-stock, and how integrated plans shift stock-outs to peripheral stores, across shelf life, penalty ratio, fleet capacity, routing cost, demand variability and issuing policy (Section 6).

Section 2 reviews related work, Section 3 defines the problem and its structural properties, Section 4 describes the policies, Section 5 the experimental design, Section 6 the results, and Section 7 concludes with managerial implications and limitations.

2. Related literature

2.1. Perishable inventory control

Perishable inventory theory studies ordering when units have a finite lifetime and the age of stock matters for issuing, outdating and service (Nahmias, 1982); Bakker et al. (2012) and Janssen et al. (2016) review the subsequent literature. Optimal policies depend on the full age vector and are computed by stochastic dynamic programming only for small instances, so practice relies on simple rules. Base-stock policies that raise the total inventory to a target are the default in automated retail ordering, and Bu et al. (2023) show that they become asymptotically optimal when lifetime, demand volume or the penalty costs grow. Stock-age information helps when lifetimes are short. Broekmeulen and van Donselaar (2009) propose to add the estimated waste of current stock to the base-stock order (EWA), Haijema (2014) studies optimal ordering, issuing and disposal decisions, Haijema and Minner (2016) compare

stock-level dependent hybrid base-stock and constant-order rules, and Haijema and Minner (2019) show in 11,177 instances that a base-stock policy with an estimated-waste correction (BSP-low-EW) performs close to the SDP optimum and better than the other practical rules. Deep reinforcement learning has been applied to perishable inventory control (De Moor et al., 2022; Boute et al., 2022; Temizöz et al., 2025) and matches strong heuristics in single-location settings (Gijsbrechts et al., 2022). Data-driven newsvendor work uses features to set perishable order quantities directly (Ban and Rudin, 2019; Huber et al., 2019). All of these studies consider a single stocking point or decouple stores; deliveries do not compete for vehicles and the cost of a delivery does not depend on which other stores are visited.

2.2. Perishable inventory routing

Inventory routing integrates replenishment with vehicle routing (Coelho et al., 2014b). For perishables, Coelho and Laporte (2014) and Alvarez et al. (2020) develop exact formulations and branch-and-cut for deterministic demand with age-dependent inventory, and Soysal et al. (2015) add demand uncertainty and emissions in a chance-constrained model. Crama et al. (2018) study stochastic perishable inventory routing and compare an expected-value policy, the deliver-up-to-level rule UL_λ, and decomposition methods; they find that the simple rule is competitive in some settings and that more advanced methods pay off in others. He et al. (2024) study a robust perishable IRP under demand uncertainty and solve robust-counterpart reformulations by Benders decomposition and an iterated local search. For non-perishable IRPs under uncertainty, Solyalı et al. (2012) derive robust counterparts, Coelho et al. (2014a) show that rolling-horizon policies using stochastic information outperform deterministic ones, and Malicki and Minner (2021) integrate dynamic safety stocks into cyclic inventory routing under non-stationary demand. Most closely related to our question, Jia et al. (2025) propose Scenario Predict-then-Optimize: quantile neural forecasts generate demand scenarios for a stochastic rolling-horizon IRP, and the single-stage scenario model performs best on synthetic and industrial data; their model has backlogging and no product ageing. Greif et al. (2024) learn the parameters of a prize-collecting TSP policy for dynamic IRP with machine learning. None of these works compares stock-age-based inventory rules, robust and scenario-based lookahead and decision-focused learning under one closed-loop simulator with perishability.

2.3. Contextual and decision-focused optimisation

When demand forecasts feed an optimisation model, one can estimate first and then optimise, or train the forecaster on the downstream cost (Elmachtoub and Grigas, 2022; Sadana et al., 2025). Decision-focused learning (DFL) differentiates through the optimisation problem, either exactly for convex problems (Amos and Kolter, 2017; Donti et al., 2017) or through surrogates and perturbations for combinatorial ones (Berthet et al., 2020; Mandi et al.,

Table 1: Positioning relative to the closest studies.

Study	Ageing	Uncertain demand	Routing	Context	Closed loop	Methods compared
Haijema and Minner (2019)	✓	✓	–	–	✓	Stock-age rules vs SDP
Broekmeulen and van Donselaar (2009)	✓	✓	–	–	✓	EWA vs base-stock
Coelho and Laporte (2014); Alvarez et al. (2020)	✓	–	✓	–	–	Exact MILP
Crama et al. (2018)	✓	✓	✓	–	✓	EV, UL_λ, decomposition
He et al. (2024)	✓	✓	✓	–	–	Robust, Benders, ILS
Solyalı et al. (2012)	–	✓	✓	–	–	Robust MILP
Jia et al. (2025)	–	✓	✓	✓	✓	Scenario PTO vs PTO, SAA, DFL
Islam Sohan and Wasi (2023)	–	✓	✓	✓	–	DFL prototype
This study	✓	✓	✓	✓	✓	Stock-age rules, UL_λ, robust, scenario PTO, DFL

2024; Tang and Khalil, 2024); solution caching reduces the number of solver calls (Mulamba et al., 2021). The theory is nuanced: under a well-specified model, estimate-then-optimise can converge faster and stochastically dominate integrated training (Hu et al., 2022; Elmachtoub et al., 2023), so gains from DFL should be expected mainly when the downstream model is a misspecified approximation. In inventory control, Alvo et al. (2023) train neural ordering policies by differentiating simulated costs through known inventory dynamics. For inventory routing, an early DFL prototype exists only as a workshop abstract without perishability (Islam Sohan and Wasi, 2023). Our method differentiates realised perishable inventory cost through a deterministic rolling-horizon PIRP whose routes are fixed by an exact MILP, so that learning corrects the safety stock embedded in the plan rather than the routing decisions.

2.4. Positioning

Table 1 positions the study. It is the first, to our knowledge, to evaluate decision-focused learning for perishable inventory routing against the inventory-control rules recommended in the perishable inventory literature and against robust and scenario-based lookahead methods, under identical information and an exact optimisation layer. The methodological novelty is the decision-focused target layer with its structural justification; the managerial contribution is evidence on when integration, stochastic lookahead and decision-focused learning are worth their cost.

3. Problem description

3.1. The periodic-review perishable inventory-routing system

A distributor replenishes a single fresh product at n retail stores $\mathcal{N} = \{1, \dots, n\}$ from a depot 0. Every morning (day t) it decides how much to deliver to each store and how to route a homogeneous fleet of K vehicles of capacity Q; each vehicle performs at most one route per day and each store is visited at most once per day. Deliveries arrive before the store opens (zero lead time), as is common for daily fresh-food replenishment. Products arrive fresh and can be sold on m consecutive days, the day of delivery included. Store i can hold at most C_i units at opening (shelf and backroom capacity).

State and stock age.. Let B_{irt} denote the units at store i at the opening of day t, before the delivery, that can still be sold on r days including day t ($r = 1, \dots, m-1$). The vector $B_t = (B_{irt})$ is the age-structured inventory state observed by the planner. Units with $r = 1$ are on their last day.

Daily dynamics.. Given deliveries $q_t = (q_{it})$ and realised demand D_{it}, the following happens at each store. (i) Available stock by remaining life is $S_{irt} = B_{irt}$ for $r < m$ and $S_{imt} = q_{it}$. (ii) Demand is served first-in first-out (FIFO), oldest units first. (iii) Unmet demand is lost. (iv) Unsold units with $r = 1$ expire. (v) Remaining units are carried overnight and lose one day of remaining life. With sales $w_{irt} = \min\{S_{irt}, (D_{it} - \sum_{r'<r} w_{ir't})^+\}$, lost sales are $u_{it} = D_{it} - \sum_r w_{irt}$, expired units are $e_{it} = S_{i1t} - w_{i1t}$, and the next state is $B_{i,r,t+1} = S_{i,r+1,t} - w_{i,r+1,t}$ for $r = 1, \dots, m-1$. As a robustness check we also simulate partial last-in first-out (LIFO) issuing, in which a fraction of customers picks the freshest units (Haijema, 2014).

Costs.. Day t costs

$$c_t = c^{\mathrm{r}}\,\mathrm{TSP}(\mathrm{routes}_t) + h\sum_{i,r} B_{i,r,t+1} + p_s \sum_i e_{it} + p_u \sum_i u_{it}, \tag{1}$$

where $\mathrm{TSP}(\cdot)$ is the travelled distance, h the holding cost per unit carried overnight, p_s the outdating (waste) cost per expired unit and p_u the lost-sales penalty. Throughout, $h \le p_s < p_u$: carrying a unit one more night is cheaper than discarding it, and a lost sale is the most expensive event.

Information and policies.. Before deciding on day t the planner observes the state B_t, the demand history up to day $t-1$, and contextual information about the next days: calendar, announced promotions and holidays, and a weather forecast. A *policy* maps this information to deliveries and routes. The objective is to minimise the expected cost $\mathbb{E}\left[\sum_{t=1}^{T} c_t\right]$ over a

horizon of T days. Delivery decisions couple days through stock age and stores through vehicle capacity and route costs; this is what distinguishes the problem from single-store perishable inventory control (Nahmias, 1982; Haijema and Minner, 2019) and makes it a stochastic perishable inventory-routing problem (Crama et al., 2018).

3.2. Lookahead model

All integrated policies in Section 4 solve, every day, an H-day lookahead model and implement only its first day (rolling horizon). The model takes the observed state B_t and one or several planning demand paths $\tilde{d}^s = (\tilde{d}^s_{ik})_{i\in\mathcal{N},k=0..H-1}$, $s \in \mathcal{S}$, as inputs. Its routing part uses a vehicle-assignment formulation with exact route costs. Let $\mathcal{R} = 2^{\mathcal{N}}$ be the set of store subsets and $\mathrm{TSP}(R)$ the length of an optimal depot-to-depot tour through R, precomputed by the Held–Karp recursion (Held and Karp, 1962). Binary z_{ivk} assigns store i to vehicle v on lookahead day k; continuous $y_{Rvk} \geq 0$ selects the route of vehicle v; $g_{ivk} \geq 0$ is the load for store i on vehicle v; $q_{ik} = \sum_v g_{ivk}$.

$$\min \sum_{k,v,R} c^{\mathrm{r}}\mathrm{TSP}(R)\, y_{Rvk} + \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\sum_{k,i}\Big(h\sum_{r\geq 2}(S^s_{irk} - w^s_{irk} - \zeta^s_{irk}) + p_s\Big(S^s_{i1k} - w^s_{i1k}\Big) + p_s\sum_{r\geq 2}\zeta^s_{irk} + p_u u^s_{ik}\Big) \tag{2}$$

$$\text{s.t.}\quad \textstyle\sum_R y_{Rvk} = 1, \quad \sum_{R\ni i} y_{Rvk} = z_{ivk}, \quad \sum_v z_{ivk} \leq 1 \qquad \forall i, v, k, \tag{3}$$

$$g_{ivk} \leq \min\{Q, C_i\}\, z_{ivk}, \quad \textstyle\sum_i g_{ivk} \leq Q \qquad \forall i, v, k, \tag{4}$$

$$S^s_{imk} = q_{ik}, \quad S^s_{ir0} = B_{irt}\ (r < m) \qquad \forall i, r, k, s,$$

$$S^s_{i,r,k+1} = S^s_{i,r+1,k} - w^s_{i,r+1,k} - \zeta^s_{i,r+1,k} \qquad \forall i, r, k, s, \tag{5}$$

$$\textstyle\sum_r w^s_{irk} + u^s_{ik} = \tilde{d}^s_{ik}, \quad w^s_{irk} + \zeta^s_{irk} \leq S^s_{irk}, \quad \sum_r (S^s_{irk} - \zeta^s_{irk}) \leq C_i \qquad \forall i, r, k, s, \tag{6}$$

with all variables non-negative and z binary. Constraint (3) lets each vehicle choose one route (the empty set has zero cost) whose store set equals the stores assigned to it; (4) enforces vehicle and shelf capacity; (5) is the age-structured balance; (6) links sales, lost sales and the shelf capacity at opening. ζ is an excess-disposal slack priced at p_s; it is only needed when $|\mathcal{S}| > 1$ (in a deterministic plan stock never exceeds what was planned) and it is never used on day 0, where the state is known. With $|\mathcal{S}| = 1$ the model is a deterministic rolling-horizon PIRP; with $|\mathcal{S}| > 1$ it is the single-stage scenario model (one routing and delivery plan for all scenarios) that Jia et al. (2025) found to perform best. A symmetry-breaking constraint assigns the lowest-indexed visited store to vehicle 1.

Proposition 1 (Exact routing cost)**.** *For every binary z satisfying (3), the only feasible y sets $y_{R_{vk},v,k} = 1$ for $R_{vk} = \{i : z_{ivk} = 1\}$. Hence the routing term of (2) equals the exact*

travelled distance of the assigned routes, and the model is an exact MILP formulation of the H-day lookahead PIRP in which only the nKH assignment variables are integer.

The proof is in Appendix A. Proposition 1 matters computationally: with ten stores the model has $2^{10}KH$ continuous route columns but only nKH binaries, whereas the arc-based vehicle-indexed formulation used in the original submission could not be solved within its time budget. Because only day 0 of each plan is implemented, the experiments impose integrality on the day-0 assignment and relax it on later lookahead days; the relaxed routing term is then a convex lower estimate of future travel cost. With this choice HiGHS (Huangfu and Hall, 2018) proves a gap of at most 0.5% for every decision within the time limit (Section 6.5); in a pilot on four episodes, imposing integrality on all lookahead days changed episode costs by between -1.5% and $+3.3\%$ at three to five times the solution time and with time-limit stops. Day-0 routes of every implemented decision are additionally re-optimised by exact enumeration of all feasible route sets.

3.3. Structural properties

The results below connect the lookahead model with the physical FIFO system and justify the learning signal used in Section 4.4. Fix one store, a delivery plan $q = (q_0, \dots, q_{H-1})$, an initial state B and a demand path d. Let $C_{\mathrm{FIFO}}(q; d)$ be the inventory cost (holding, waste and lost sales) of the physical system above, and let $V_{\mathrm{LP}}(q; d)$ be the optimal value of the *recourse LP*: constraints (5)–(6) for one scenario with q fixed, the issuing order left free and early disposal ζ allowed.

Theorem 1 (Complete recourse and FIFO optimality)**.** *(i) For any state with $\sum_r B_{ir} \le C_i$ and any non-negative planning demand, the lookahead MILP is feasible and its optimal value is finite. (ii) For fixed deliveries, the recourse LP has an optimal solution in which every day's sales follow FIFO and demand is served whenever stock is available. (iii) $V_{\mathrm{LP}}(\cdot; d)$ is convex and piecewise linear, and $V_{\mathrm{LP}}(q; d) \le C_{\mathrm{FIFO}}(q; d)$, with equality whenever early disposal is not profitable along the FIFO trajectory.*

Part (ii) corrects a claim of the original submission, which stated that age-increasing holding costs make every optimal solution FIFO without a disposal option. That statement is false: if an expiring unit and a fresh unit are on the shelf and only one more unit will ever be sold, selling the fresh unit and letting the old one expire saves the fresh unit's holding cost. Allowing disposal at cost p_s restores the exchange argument (the younger unit can always replicate the older unit's path) and makes the recourse a convex relaxation of the physical system.

Proposition 2 (Marginal value of a delivery). *(a) The dual multiplier π_k of the fresh-stock constraint $S_{imk} = q_k$ is a subgradient of V_{LP} with respect to q_k, and $-p_u \le \pi_k \le p_s$. (b) If $h \le p_s$, $C_{\mathrm{FIFO}}(\cdot; d)$ is piecewise linear and Lipschitz, and its one-sided partial derivatives satisfy*

$$-p_u \;\le\; \frac{\partial C_{\mathrm{FIFO}}}{\partial q_k} \;\le\; p_s + (H - 1 - k)\, h. \tag{7}$$

Proposition 2 states the familiar newsvendor logic in a multi-period, age-structured form: an extra delivered unit can at most prevent one lost sale, and at worst it is carried to the end of the horizon or wasted (possibly by displacing a later delivery). The asymmetry $p_u \gg p_s$ is what makes point forecasts a poor input for perishable replenishment. The bound in (b) is tight (Section 6.6).

The learned planning targets of Section 4.4 enter the lookahead through the deliveries they induce. For a fixed visit pattern, a delivery on day k covers the planned demand of the days until the next visit, limited by shelf life and horizon, net of the stock that will still be usable. Let W_k denote this coverage window.

Proposition 3 (Decision-focused stationarity). *Fix the visit pattern and consider planning targets $\tilde{d}_j$ at which no capacity binds and the coverage map is differentiable. Then $\partial q_k / \partial \tilde{d}_j \in \{0, 1\}$, with value one exactly when day j lies in the coverage window W_k and is not already covered by planned stock, and for such j*

$$\frac{\partial}{\partial \tilde{d}_j}\, \mathbb{E}\Big[C_{\mathrm{FIFO}}(q(\tilde{d}); D) \mid \xi\Big] = \mathbb{E}\Big[\frac{\partial C_{\mathrm{FIFO}}}{\partial q_{k(j)}} \,\Big|\, \xi\Big], \tag{8}$$

where $k(j)$ is the visit covering day j. A stationary point of the decision-focused objective therefore satisfies the first-order optimality condition of the stochastic program in which only the delivery quantities of the fixed visit pattern are optimised. With a single-day shelf life, (8) vanishes exactly at the perishable newsvendor quantile, $F_{D|\xi}(\tilde{d}) = p_u/(p_u + p_s)$.

Proposition 3 clarifies what decision-focused training can and cannot do. It can correct the safety stock embedded in deterministic plans, store by store and context by context, which is where the cost asymmetry and stock age act; it does not optimise visit days, which are left to the MILP. It also explains the relation to the estimate-then-optimise literature: when the forecasting model is well specified and decisions are optimised against the full predictive distribution, integrated training has no first-order advantage (Hu et al., 2022; Elmachtoub et al., 2023); a gain is possible only when the downstream model is a misspecified deterministic approximation, as all rolling-horizon PIRP policies are.

4. Replenishment policies

All policies receive the same information each day: the observed age-structured stock B_t and the same contextual forecasts. They differ only in how the forecasts and the stock are turned into deliveries and routes. Table 2 summarises the nine policies; parameters marked as tuned are selected on validation episodes only.

4.1. Shared forecasts

For every store i, planning day t and lookahead offset $k \in \{0, \dots, H-1\}$, one LightGBM model (Ke et al., 2017) predicts the mean $\hat{\mu}_{i,t+k}$ and eleven LightGBM quantile models predict the quantiles $\hat{Q}_\tau(D_{i,t+k} \mid \xi)$ for $\tau \in \{0.05, 0.1, 0.2, \dots, 0.9, 0.95\}$ (sorted to avoid crossing). Features are store identity, offset k, calendar, announced discount and holiday, temperature forecast, and summaries of the last 1, 7 and 28 days of demand. A forecast scale is $\hat{\sigma} = (\hat{Q}_{0.9} - \hat{Q}_{0.1})/(2z_{0.9})$, and quantiles at other levels are linearly interpolated. Quantile regression of this type is the standard data-driven newsvendor input (Ban and Rudin, 2019; Huber et al., 2019).

4.2. Inventory-control rules with route-second delivery

These policies represent current replenishment practice: an inventory rule sets the store orders, and the orders are then routed. Deliveries are routed by an exact capacitated vehicle-routing oracle that enumerates all feasible route sets (possible because $n = 10$). If the fleet cannot carry all orders, the part exceeding today's median demand is reduced proportionally until a feasible packing exists.

BSP: periodic base-stock.. Stores are served on a fixed cycle of $R \in \{1, 2\}$ days (for $R = 2$ in two alternating groups). On a delivery day, store i is raised to the order-up-to level $S_{it} = \hat{Q}_\alpha\Big(\sum_{k<R} D_{i,t+k}\Big)$ (normal approximation for $R > 1$). The inventory position counts all units regardless of age, as in most retail ordering systems (Haijema and Minner, 2019).

BSP-EW: base-stock with estimated waste.. Identical to BSP, except that units of the current stock that are expected to expire unsold during the cycle are removed from the inventory position: $q_{it} = (S_{it} - \sum_r B_{irt} + \widehat{EW}_{it})^+$, where $\widehat{EW}_{it}$ is computed by projecting the age profile forward under FIFO with the mean forecasts. This is the "estimated withdrawal and aging" idea of Broekmeulen and van Donselaar (2009); waste-anticipating base-stock rules of this family were found to be near-optimal among practical stock-age-dependent policies by Haijema and Minner (2019).

UL_λ: deliver-up-to-level rule (Crama et al., 2018).. A store is visited when its stock does not cover the current day's demand at service level β, i.e. when $\sum_r B_{irt} < \hat{Q}_\beta(D_{it})$; it then receives enough to reach the α-quantile of the demand of the next λ days, net of stock expected to expire. This is the state-dependent "deliver-up-to-level" rule that Crama et al. (2018) identify as a simple policy with good performance in some settings; the original paper combines it with its own routing heuristic, whereas here the selected deliveries are routed exactly.

4.3. Integrated rolling-horizon policies

These policies solve the lookahead MILP of Section 3.2 every day with $H = 3$ ($H = 2$ when $m = 2$) and implement day 0. They differ only in the planning demand.

- **LA-Mean** uses the point forecasts $\hat{\mu}$: the classical predict-then-optimise policy.
- **LA-Quantile** uses $\hat{Q}_\alpha$ for every store and day. With α tuned on validation episodes, this is a parametric cost-function approximation in the sense of Powell (2019): a deterministic lookahead whose inputs are adjusted to perform well under uncertainty.
- **LA-Robust** uses budgeted demand uncertainty (Bertsimas and Thiele, 2006). Demand on lookahead day k ranges over $[\hat{\mu}_k, \hat{\mu}_k + \hat{\delta}_k]$ with $\hat{\delta}_k = \hat{Q}_{0.95} - \hat{\mu}_k$, and the cumulative scaled deviation of the first $k+1$ days is at most $\Gamma_k = \min\{\Gamma\sqrt{k+1}, k+1\}$. The planning path reproduces the worst-case cumulative demand of every prefix, which is the deterministic equivalent shown by Bertsimas and Thiele (2006) for inventory systems and used in robust inventory routing (Solyalı et al., 2012). Γ is tuned.
- **ScenPO-SS** is the Scenario Predict-then-Optimize method of Jia et al. (2025): ten demand scenarios are sampled from the multi-horizon quantile forecasts (each scenario draws a quantile band uniformly and then draws every store-day value uniformly inside that band), and the single-stage scenario model chooses one delivery and routing plan that minimises the average cost over the scenarios. Two adaptations are needed and disclosed: inventory follows the perishable age-structured dynamics with lost sales instead of the original non-perishable backlogging model, and the quantile model is LightGBM rather than an MQ-RNN, so that all policies use the same forecasts.
- **LA-DFL** (proposed) uses decision-focused planning targets (Section 4.4).
- **PI** solves the same lookahead with the realised demand of the next H days. It is a value-of-information reference, not a lower bound.

Table 2: Policies compared. All use the same observed stock and the same forecasts.

Policy	Family	Decision rule	Tuned on validation
BSP	Inventory rule, route second	Periodic base-stock on total stock	Cycle R, quantile α
BSP-EW	Inventory rule, route second	Base-stock plus estimated waste (Broekmeulen and van Donselaar, 2009; Haijema and Minner, 2019)	R, α
UL_λ	Stochastic PIRP rule	Deliver-up-to-level (Crama et al., 2018)	λ, α, β
LA-Mean	Rolling-horizon PIRP	Mean forecasts	–
LA-Quantile	Rolling-horizon PIRP (CFA)	Quantile forecasts	α
LA-Robust	Rolling-horizon PIRP	Budgeted uncertainty (Bertsimas and Thiele, 2006)	Γ
ScenPO-SS	Rolling-horizon stochastic PIRP	Scenario PTO (Jia et al., 2025)	– (10 scenarios)
LA-DFL	Rolling-horizon PIRP	Decision-focused targets (this paper)	Network weights, early stopping
PI	Reference	Realised demand	–

4.4. Decision-focused planning targets

LA-DFL keeps the deterministic lookahead, which is fast and easy to explain to planners, and learns how much each planning target should deviate from the forecast. For store i, planning day t and offset k the target is

$$\tilde{d}_{i,t+k}(\theta) = \left(\hat{Q}_{\alpha^\star}(D_{i,t+k} \mid \xi) + \hat{\sigma}_{i,t+k}\, \delta_\theta(\phi_{itk})\right)^+, \tag{9}$$

where $\alpha^\star$ is the quantile level selected for LA-Quantile, and δ_θ is a two-layer neural network whose output layer is initialised to zero, so that training starts exactly at the tuned LA-Quantile policy. The input ϕ_{itk} contains the forecast features, the scaled mean, scale and 5–95% range of the forecast, the offset k, and the store's current stock by remaining life. The correction can therefore depend on context (e.g. promotions, which raise both mean and variance) and on stock age, which neither the forecasts nor a single quantile level capture.

Learning signal.. The visit pattern chosen by the MILP is piecewise constant in the targets, so its gradient carries no information. Given the pattern, however, deliveries are piecewise linear in the targets: each delivery covers the planned demand of its coverage window W_k, net of the usable stock projected by FIFO, capped by the remaining shelf room and scaled to

Algorithm 1 Training of decision-focused planning targets (LA-DFL)

Require: training states $\{(B^{(s)}, \xi^{(s)}, D^{(s)})\}$ from LA-Quantile runs; tuned level $\alpha^\star$; epochs E

initialise $\delta_\theta \equiv 0$; $\theta^\star \leftarrow \theta$; $J^\star \leftarrow$ validation cost of LA-Quantile
for $e = 1, \dots, E$ **do**
    for every state s: solve the lookahead MILP with targets $\tilde{d}_\theta^{(s)}$; store visit pattern $V^{(s)}$
    for each minibatch $\mathcal{B}$ **do**
        $L \leftarrow |\mathcal{B}|^{-1} \sum_{s \in \mathcal{B}} C_{\mathrm{FIFO}}\big(q(\tilde{d}_\theta^{(s)}; V^{(s)}), D^{(s)}\big)$ ▷ differentiable coverage layer + FIFO simulator
        $\theta \leftarrow \mathrm{Adam}(\theta, \nabla_\theta L)$
    end for
    $J \leftarrow$ closed-loop validation cost of LA-DFL with θ; **if** $J < J^\star$ **then** $\theta^\star \leftarrow \theta$, $J^\star \leftarrow J$
end for
return $\theta^\star$

the vehicle capacity. We implement this coverage map $q(\tilde{d}; V)$ as a differentiable layer and evaluate the realised H-day FIFO cost of the resulting deliveries under the observed demand with a differentiable simulator. Automatic differentiation then returns

$$\nabla_\theta \, C_{\mathrm{FIFO}}\Big(q(\tilde{d}_\theta; V), D\Big) = \left(\frac{\partial \tilde{d}}{\partial \theta}\right)^\top \left(\frac{\partial q}{\partial \tilde{d}}\right)^\top \nabla_q C_{\mathrm{FIFO}}, \tag{10}$$

whose last factor is the marginal value of a delivery bounded in Proposition 2, and whose stationary points are characterised in Proposition 3. In words: if, in hindsight, an extra unit in a store's delivery would have prevented a lost sale, the targets of the days that delivery covered are pushed up; if the extra unit would have been wasted or carried, they are pushed down; and nothing changes when the store was not visited or capacity was binding.

Training.. Training states are the decision epochs visited by the tuned LA-Quantile policy on training episodes. Each epoch re-solves the MILP at the current targets to refresh the visit patterns (a form of solution caching, Mulamba et al., 2021), takes Adam steps on the average realised cost of minibatches, and evaluates the closed-loop cost of the current network on the validation episodes; the network with the lowest validation cost is kept, which may be the LA-Quantile starting point. Algorithm 1 summarises the procedure.

Compared with the decision-focused learner of the original submission, three changes are essential. The learner no longer differentiates a heuristic surrogate of a MILP that was in practice never solved; the MILP is solved to a proven gap and only supplies visit patterns. The loss is the realised physical cost rather than regret against a heuristic full-information plan. And training and model selection are performed on the closed-loop system that is evaluated.

5. Experimental design

5.1. Network, products and costs

The network consists of a depot and the first ten customers of Solomon's R101 instance (Solomon, 1987); coordinates define Euclidean distances and time windows are ignored. Two vehicles serve the stores. Vehicle capacity is set through the fleet ratio $\rho = KQ/\bar{D}$, where $\bar{D}$ is mean total daily demand; the base case has $\rho = 1.25$ (119 units per vehicle), so that the fleet can carry a quarter more than average demand, as is common in grocery distribution where vehicles are sized for peak days. Shelf capacity is three days of mean store demand. The base product has a shelf life of $m = 3$ days. Costs are $h = 1$ per unit and night, $p_s = 5$ per expired unit and $p_u = 20$ per lost sale, with routing cost one per distance unit; per unit of demand, waste is five times and a lost sale twenty times as expensive as one night of holding, the asymmetry studied in the original submission.

5.2. Demand and information

Demand is generated by a contextual negative-binomial model (code in the replication package). Store base levels range from 10 to 22 units per day; demand varies with the day of the week (Saturday +30%), announced promotions (12% of store-days, discounts of 10–30%, multiplying demand by up to 1.8 and lowering the negative-binomial size parameter from 8 to 3.2, so that promotions raise both level and variability), temperature, holidays, and a latent store level following a log-AR(1) process with autocorrelation 0.92 that can only be inferred from recent sales. The coefficient of variation of daily store demand is about 0.4 on regular days. The temperature forecast error grows with the lead time. The demand model is hand-specified; it is designed to reproduce qualitative features of fresh retail demand (weekly pattern, promotion lifts, heteroscedasticity, persistence), not calibrated to a specific retailer, and no results should be read as estimates for a particular chain.

Each experiment uses 100 training, 20 validation and 40 test episodes, generated with disjoint random seeds. An episode consists of 35 days of history (to compute lagged features), 21 decision days, and four further days so that lookahead forecasts and the perfect-information reference are defined on the last decision days. Every policy starts each test episode with empty stores and is evaluated on the same 40 demand paths (common random numbers).

5.3. Configurations

Besides the base case we vary one factor at a time: shelf life $m \in \{2, 4\}$; lost-sales penalty $p_u \in \{10, 40\}$; fleet ratio $\rho \in \{1.10, 1.60\}$; routing cost $c^{\mathrm{r}} = 3$ per distance unit, which makes consolidation of deliveries more valuable; demand variability (negative-binomial size 3 instead of 8, i.e. coefficient of variation about 0.6 on regular days); and partial LIFO issuing, where

30% of customers take the freshest units. In the LIFO configuration all policies are tuned and trained under FIFO and evaluated under mixed issuing, which tests robustness to a misspecified issuing assumption that practitioners often face.

5.4. Tuning, training and inference

All parameters are selected on validation episodes by closed-loop simulation: BSP and BSP-EW over $R \in \{1, 2\}$ and $\alpha \in \{0.6, 0.7, 0.8, 0.9, 0.95\}$; UL_λ over $\lambda \in \{1, 2, 3\}$, $\alpha \in \{0.7, 0.8, 0.9, 0.95\}$, $\beta \in \{0.5, 0.7, 0.9\}$; LA-Quantile over $\alpha \in \{0.5, \ldots, 0.9\}$; LA-Robust over $\Gamma \in \{0.25, 0.5, 1, 1.5\}$. LA-DFL is trained on the decision states visited by the tuned LA-Quantile policy in 60 training episodes (1,260 states), with eight epochs, visit patterns re-optimised every second epoch, Adam (learning rate $2 \cdot 10^{-3}$, minibatch 64), a two-layer network with 64 SiLU units, and validation-based model selection. Test episodes are used once.

The prespecified primary hypotheses compare LA-DFL with LA-Quantile, ScenPO-SS, LA-Robust, BSP-EW and UL_λ in the base case. The unit of analysis is the episode; for each comparison we report the mean paired cost difference, a 95% paired bootstrap interval (10,000 resamples), paired t and Wilcoxon signed-rank tests, and Holm-adjusted p-values over the five primary comparisons (Holm, 1979). All other comparisons and configurations are reported with paired intervals as secondary evidence.

5.5. Optimisation settings

Lookahead MILPs are solved by HiGHS through SciPy with a relative gap of 0.5% and time limits of 20 seconds (deterministic) and 60 seconds (scenario), with integer vehicle assignments on the implemented day and relaxed assignments on later lookahead days (Section 3.2). The DFL oracle uses the same model with a 1% gap. If a time limit were reached the best incumbent would be used and its proven gap recorded; Section 6.5 reports solver statistics. All experiments run on a 16-core desktop computer with 15 parallel worker processes; the code, seeds and configuration files are in the replication package.

6. Results

6.1. Base case

Table 3 reports test-set averages over the 40 episodes and Figure 1 decomposes the costs. Tuning selected order-up-to levels at the 0.95 quantile with daily delivery for BSP and BSP-EW, UL_λ with $\lambda = 1$, $\alpha = 0.95$, $\beta = 0.9$, quantile level $\alpha^\star = 0.9$ for LA-Quantile and budget $\Gamma = 1.0$ for LA-Robust.

Table 3: Base case: mean cost per 21-day test episode and its components, fill rate, waste rate (expired units per delivered unit) and store visits per day (40 episodes).

Policy	Total cost (SD)	Routing	Holding	Waste	Lost sales	Fill rate (%)	Waste rate (%)	Visits/day
BSP	10,472 (2,012)	4,293	2,927	475	2,777	96.5	2.55	9.5
BSP-EW	10,515 (1,989)	4,311	2,973	478	2,752	96.5	2.57	9.6
UL_λ	10,452 (2,061)	4,240	2,935	491	2,784	96.5	2.64	9.2
LA-Mean	18,670 (2,772)	3,935	805	30	13,900	81.6	0.20	9.7
LA-Quantile	11,768 (2,273)	3,538	3,868	1,098	3,264	95.8	5.44	7.9
LA-Robust	11,451 (2,150)	3,740	3,712	887	3,113	96.0	4.49	8.6
ScenPO-SS	11,527 (1,732)	3,419	3,832	1,063	3,213	95.9	5.35	7.7
LA-DFL	11,085 (2,187)	3,820	2,995	532	3,738	95.2	2.76	8.9
PI	5,298 (2,204)	4,133	97	0	1,068	98.8	0.00	9.9

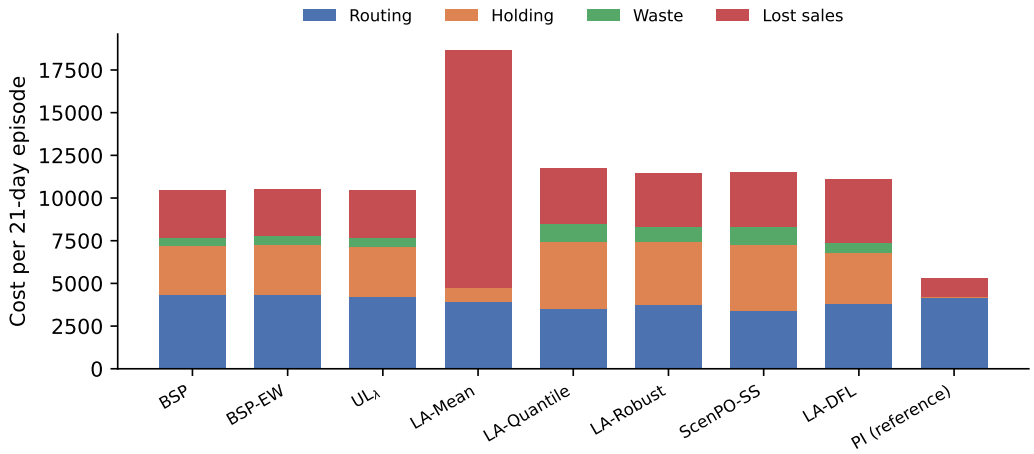


Figure 1: Cost composition of each policy in the base case.

Four observations stand out. First, uncertainty is expensive: with perfect demand information the same lookahead costs 5,298, about half of the best implementable policy, almost entirely because lost sales and holding vanish. Second, point forecasts are a poor input for perishable replenishment. LA-Mean plans to sell exactly the expected demand and loses 13,900 per episode in lost sales (fill rate 81.6%), as Proposition 2 suggests when p_u is four times p_s. Third, the three stock-age-based inventory rules are the cheapest policies: UL_λ, BSP and BSP-EW cost between 10,452 and 10,515 with a fill rate of 96.5% and a waste rate near 2.6%. With daily review, little of the current stock is at risk of expiring before it is sold, so the waste correction of BSP-EW rarely changes an order.

Fourth, integration with routing does reduce travel but loses more on the inventory side. The lookahead policies visit 7.7–8.9 stores per day instead of 9.2–9.6 and save 400–900 in routing, but LA-Quantile, LA-Robust and ScenPO-SS incur 27–32% more holding cost than the rules and waste about twice as much (waste rates 4.5–5.4%). A deterministic plan that covers two or three days with quantile targets stacks a safety buffer on every covered day, even

Table 4: Paired episode-level comparisons of LA-DFL with every other policy (base case). Negative differences favour LA-DFL. Holm adjustment over the five prespecified primary comparisons.

Comparison	Mean diff.	95% bootstrap CI	Rel. (%)	DFL wins (%)	p (t)	p (Wilcoxon)	Holm p
LA-DFL − BSP	612.5	[407.1, 819.0]	5.9	15	<0.001	<0.001	–
LA-DFL − BSP-EW	569.4	[363.2, 778.4]	5.4	22	<0.001	<0.001	<0.001
LA-DFL − UL_λ	633.3	[433.5, 836.9]	6.1	18	<0.001	<0.001	<0.001
LA-DFL − LA-Mean	-7,584.7	[-8,132.6, -7,040.4]	-40.5	100	<0.001	<0.001	–
LA-DFL − LA-Quantile	-683.6	[-919.6, -465.0]	-5.6	78	<0.001	<0.001	<0.001
LA-DFL − LA-Robust	-366.0	[-526.6, -204.1]	-3.2	80	<0.001	<0.001	<0.001
LA-DFL − ScenPO-SS	-441.8	[-721.9, -154.4]	-4.1	70	0.005	0.003	0.005
LA-DFL − PI	5,787.2	[5,453.6, 6,140.5]	118.1	0	<0.001	<0.001	–

though the next day's re-planning will correct any shortfall; the stochastic scenario model reduces lost sales but, because its single plan must work for all scenarios, it also covers several days generously.

LA-DFL is the cheapest of the integrated policies (Table 4). It reduces cost by -5.6% relative to LA-Quantile (95% CI [-920, -465], lower in 78% of the episodes), by -3.2% relative to LA-Robust and by -4.1% relative to ScenPO-SS (Holm-adjusted $p \leq 0.005$). It achieves this by halving the waste rate of LA-Quantile (2.76% vs. 5.44%) and cutting its holding cost by more than a fifth, while accepting somewhat more lost sales. It does not, however, beat the inventory rules: LA-DFL costs 5.4% more than BSP-EW and 6.1% more than UL_λ, and both differences are significant. The rules remain better in 78–85% of the episodes.

6.2. What the decision-focused targets learn

Table 5 summarises the learned corrections on the decision states that LA-DFL visits in 20 test episodes. The dominant pattern is temporal. For the implemented day the network keeps the target at about the 0.91 quantile, essentially the tuned LA-Quantile level, but it lowers the targets of the second and third lookahead days to about the 0.79 and 0.76 quantiles. In inventory terms, the network has learned that only today's shortfall is irreversible: tomorrow's planned demand will be re-planned with tomorrow's stock and forecast, so a full safety buffer on that day mostly turns into carried stock and, with a three-day shelf life, into waste. This is precisely the recourse that a deterministic lookahead ignores and that Proposition 3 lets the gradient account for. Context-specific corrections are smaller: targets are lowered somewhat more on promotion days, where the forecast spread is already wide, and slightly less when the store holds more or older stock. Validation cost improved during the first three epochs and deteriorated afterwards (Figure 2), which is why model selection on closed-loop validation cost is part of the method.

Table 5: Learned corrections of LA-DFL on the decision states of 20 test episodes (base case): mean correction δ_θ in forecast-scale units, implied quantile level of the planning target, and target relative to the mean forecast. k is the lookahead offset; stock is measured in days of mean store demand.

Targets	n	Correction δ_θ (SD)	Implied quantile level	Target / mean forecast
all	12,600	-0.40 (0.38)	0.821	1.44
k=0	4,200	+0.08 (0.16)	0.907	1.65
k=1	4,200	-0.57 (0.17)	0.792	1.36
k=2	4,200	-0.69 (0.18)	0.764	1.30
no promotion	11,078	-0.37 (0.37)	0.825	1.44
promotion	1,522	-0.61 (0.38)	0.788	1.43
stock <0.25 d	2,826	-0.45 (0.38)	0.812	1.40
stock 0.25-0.75 d	4,323	-0.43 (0.38)	0.815	1.42
stock >0.75 d	5,451	-0.35 (0.37)	0.831	1.46
no expiring units	8,256	-0.42 (0.38)	0.816	1.42
units expiring today	4,344	-0.35 (0.37)	0.830	1.47

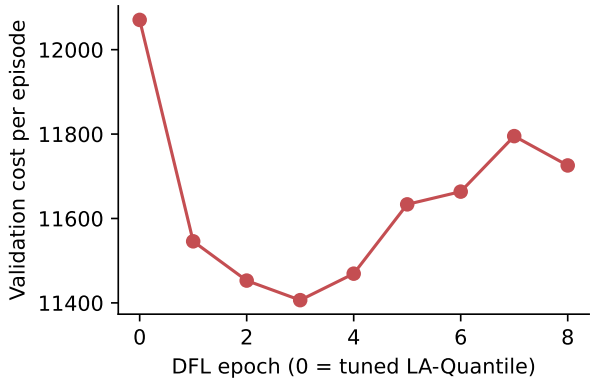


Figure 2: Closed-loop validation cost of LA-DFL by training epoch (base case). Epoch 0 is the tuned LA-Quantile policy.

6.3. Sensitivity analysis

Table 6 reports test costs across the configurations, and Table 7 the paired differences of LA-DFL. Four patterns emerge across the nine variations of the base case. (i) Stock-age-based rules with daily delivery are the cheapest policies in every configuration except the one with expensive routing: UL_λ (BSP when $m = 2$) is 2.4–8.9% cheaper than the best lookahead policy. The advantage is largest when lost sales are expensive ($p_u = 40$) and smallest when they are cheap ($p_u = 10$), in line with the high service quantile at which the rules operate. (ii) With three times higher routing cost, integration pays: ScenPO-SS is the cheapest policy, 2.3% below UL_λ and 7.1% below BSP-EW, and every uncertainty-aware lookahead beats the daily base-stock rules, which can no longer afford to visit every store every day. (iii) No lookahead policy dominates. LA-DFL is the best lookahead in five configurations (base, $m = 4$, $p_u = 10$, high variability, partial LIFO), ScenPO-SS in four ($m = 2$, $p_u = 40$, $\rho = 1.10$, $c^{\text{r}} = 3$) and

Table 6: Mean test cost per episode by configuration (one factor changed at a time; bold: lowest implementable policy), and fill rates of selected policies.

Policy	Base	$m = 2$	$m = 4$	$p_u = 10$	$p_u = 40$	$\rho = 1.10$	$\rho = 1.60$	High var.	$c^{\mathrm{r}} = 3$	30% LIFO
BSP	10,472	**12,204**	10,182	8,837	12,842	11,525	9,674	14,934	19,058	10,816
BSP-EW	10,515	12,241	10,186	8,862	12,892	11,485	9,732	15,002	19,138	10,882
UL_λ	**10,452**	12,318	**9,986**	**8,713**	**12,797**	**11,462**	**9,631**	**14,825**	18,195	**10,774**
LA-Mean	18,670	19,135	18,635	11,714	32,633	18,678	18,670	25,633	23,901	18,690
LA-Quantile	11,768	13,542	10,869	9,151	14,895	12,721	10,210	16,697	18,938	12,220
LA-Robust	11,451	13,191	10,815	8,972	14,478	12,581	9,968	16,037	18,673	11,924
ScenPO-SS	11,527	12,870	10,754	9,184	14,051	11,906	10,713	16,269	**17,787**	11,920
LA-DFL	11,085	13,688	10,681	8,924	14,883	12,470	10,143	15,971	18,462	11,398
PI (reference)	5,298	5,518	5,298	4,758	6,364	6,601	4,584	6,763	13,268	5,296
Fill rate (%), LA-DFL	95.2	93.0	95.4	93.3	95.9	93.6	95.9	92.3	95.1	95.2
Fill rate (%), ScenPO-SS	95.9	95.5	95.9	92.4	97.5	94.7	96.2	92.1	94.6	95.8
Fill rate (%), LA-Quantile	95.8	91.4	96.1	92.5	95.9	92.1	96.1	88.3	93.0	95.7
Fill rate (%), BSP-EW	96.5	94.8	97.3	94.9	97.3	95.2	97.4	93.7	96.5	96.5

Table 7: Paired cost difference of LA-DFL relative to each comparator, in percent of the comparator's cost, by configuration. Bold: 95% paired bootstrap interval excludes zero.

LA-DFL	Base	$m = 2$	$m = 4$	$p_u = 10$	$p_u = 40$	$\rho = 1.10$	$\rho = 1.60$	High var.	$c^{\mathrm{r}} = 3$	30% LIFO
vs LA-Quantile	**-5.6**	+0.8	**-1.6**	**-2.5**	-0.7	**-2.2**	-0.6	**-4.1**	**-2.5**	**-6.6**
vs ScenPO-SS	**-4.1**	**+5.6**	-0.6	**-2.8**	**+5.3**	**+4.4**	**-5.3**	-1.7	**+4.0**	**-4.6**
vs LA-Robust	**-3.2**	**+3.5**	**-1.1**	-0.6	**+2.6**	-1.0	**+1.7**	-0.2	-1.0	**-4.4**
vs BSP-EW	**+5.4**	**+11.3**	**+4.9**	+0.7	**+15.5**	**+8.6**	**+4.1**	**+6.6**	**-3.5**	**+4.6**
vs UL_λ	**+6.1**	**+10.6**	**+7.1**	**+2.4**	**+16.1**	**+8.7**	**+5.1**	**+8.0**	**+1.7**	**+5.8**

LA-Robust in one ($\rho = 1.60$), where ScenPO-SS is the weakest uncertainty-aware lookahead; LA-Mean is the most expensive policy in all ten. (iv) The decision-focused correction never makes the deterministic lookahead significantly worse. LA-DFL is 0.6–6.6% cheaper than LA-Quantile in nine configurations (significant after Holm adjustment in five: base, $p_u = 10$, high variability, $c^{\mathrm{r}} = 3$ and partial LIFO) and 0.8% more expensive, not significantly, with $m = 2$. The direction of the correction adapts to the setting: it lowers stock where the quantile plan over-stocks (waste rate 2.8% vs. 5.4% in the base case, 4.5% vs. 7.8% under partial LIFO) and raises it where the plan under-stocks (fill rate 92.3% vs. 88.3% with high variability, 95.1% vs. 93.0% with expensive routing). Finally, partial LIFO issuing raises the cost of the rules and of the uncertainty-aware lookahead policies by 2.8–4.1% (LA-DFL being least affected) but leaves the cheapest policy and the best lookahead unchanged, so the FIFO assumption used in planning is not critical here.

6.4. Store-level service

Average fill rates hide how service is distributed over the network (Figure 4). The inventory rules serve every store at least 95.4% and, if anything, serve distant stores slightly better (correlation between store fill rate and depot distance +0.75 for BSP-EW). The deterministic lookahead policies do the opposite: their fill rate falls with distance (correlations between

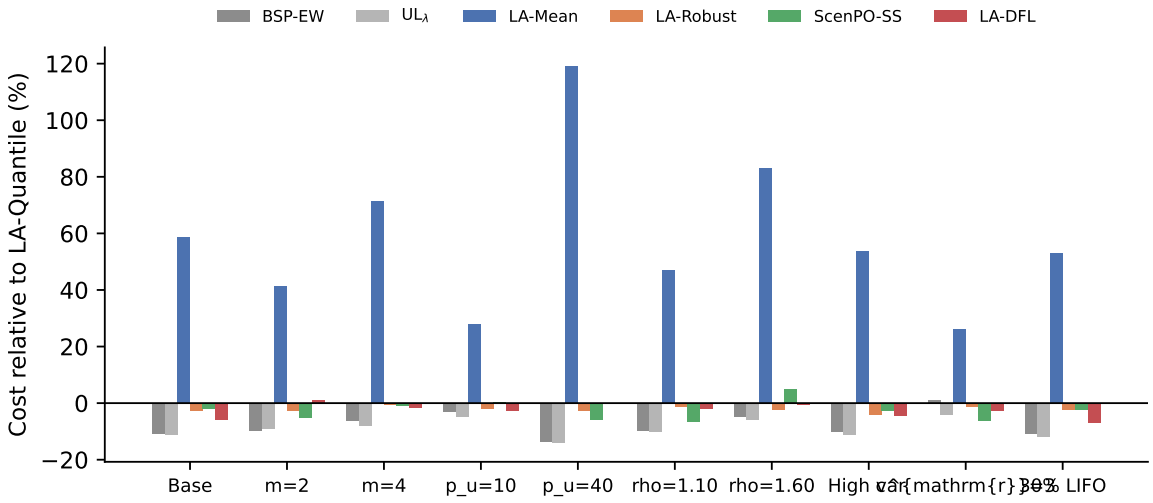


Figure 3: Cost of selected policies relative to LA-Quantile, by configuration.

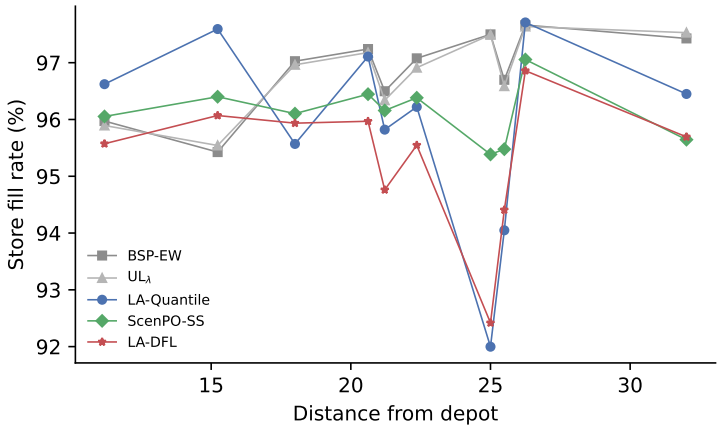


Figure 4: Store fill rate against distance from the depot (base case, mean over 40 test episodes).

−0.19 and −0.36), and the worst-served store, a mid-distance store, reaches only 92.0% under LA-Quantile and 92.4% under LA-DFL. Consolidating deliveries saves the most travel at stores that are expensive to reach, and the plan then accepts shortages there. ScenPO-SS spreads service more evenly (worst store 95.4%). This effect is relevant for retailers with service-level agreements per store: an integrated plan that is cheaper overall may concentrate stock-outs at peripheral stores.

6.5. Computation

All 5,040 lookahead decisions of the base-case test were solved to a proven gap of at most 0.50%, with 0 time-limit stops and 0 fallbacks; the same holds in every configuration reported. A deterministic lookahead decision takes 1.1–1.4 seconds on average and a ScenPO-SS decision

Table 8: Numerical verification of the structural results.

Result	Check	Outcome
Theorem 1(i)	60 lookahead MILPs ($m \in \{2,3,4\}$, 1 or 5 scenarios) with demand from zero to four times fleet capacity	all feasible, all solved to proven gap
Theorem 1(ii)	3,000 random recourse instances ($m \in \{2,3,4\}$, $H \in \{2,\dots,5\}$, random costs); FIFO- and greedy-constrained MILP value vs. V_{LP}	max. absolute difference 10^{-6}
Theorem 1(iii)	$V_{\mathrm{LP}} \le C_{\mathrm{FIFO}}$ in the same instances	holds in 3,000/3,000; equality in 63%
Proposition 2(a)	dual bounds $[-p_u, p_s]$; subgradient inequality in 9,000 random directions	3,000/3,000; 9,000/9,000
Proposition 2(b)	autograd derivative of C_{FIFO} within (7)	3,000/3,000; bound attained in 71%
Proposition 3	$m = 1$, negative-binomial demand (mean 15), SGD on the decision-focused loss	learned target 20.03; empirical 0.8-quantile 20.0; $F(\tilde{d}) = 0.809$ vs. $p_u/(p_u+p_s) = 0.8$
Coverage layer	40 lookahead MILPs; deliveries implied by the visit pattern vs. MILP deliveries	98.7% of delivered volume reproduced (mean relative L_1 deviation 1.3%)

3.01 seconds; the rules need milliseconds. Training LA-DFL (1,260 states, eight epochs including closed-loop validation) took about 16 minutes with 15 worker processes, and one base-case configuration, including tuning of all policies, about 35 minutes. Once trained, LA-DFL costs no more online than any deterministic lookahead.

6.6. Verification of the structural results

Table 8 reports numerical checks of the results in Section 3.3 (script `verify_theory.py` in the replication package). Theorem 1(ii) is checked by solving the recourse a second time with FIFO issuing and greedy service imposed through binary indicator constraints: if the theorem were false, the constrained value would exceed V_{LP} in some instance. The bound of Proposition 2(b) is attained in 71% of the random instances, so it cannot be tightened in general. The physical FIFO cost equals V_{LP} in 63% of the random instances; the remaining gap is the value of discarding stock early, which a store does not do in practice. The original submission's claim that age-increasing holding costs alone make every optimal recourse solution FIFO is contradicted by the example given after Theorem 1.

7. Discussion and conclusion

7.1. Managerial implications

The study asked how contextual forecasts should be turned into replenishment decisions when perishable stock ages and deliveries share vehicles. Five implications follow for planners of fresh-food distribution.

1. **Benchmark against simple stock-age-based rules before integrating.** Daily base-stock replenishment at a high service quantile, with the resulting orders routed optimally, was the cheapest policy in nine of the ten settings, by 2.4–8.9% over the best integrated policy, including longer and shorter shelf lives, lower and higher lost-sales penalties, tighter and looser fleets, higher demand variability and partial LIFO issuing. In these settings the travel saved by consolidating deliveries is smaller than the extra holding, waste and lost sales that consolidation causes. This extends the finding of Haijema and Minner (2019) and Crama et al. (2018) that well-tuned simple rules are hard to beat to a setting with shared routes and contextual forecasts.

2. **Integration pays when routing is expensive, and then uncertainty must be modelled explicitly.** With three times higher travel cost, the scenario-based lookahead of Jia et al. (2025) was the cheapest policy and every uncertainty-aware lookahead beat daily base-stock. Deterministic lookahead with mean forecasts was the worst policy in every setting; quantile targets repair much of this. Among the uncertainty-aware lookahead methods none dominated: the scenario lookahead was best in four settings, including those with tight capacity or expensive routing, but the weakest of them when the fleet had ample slack.

3. **If a deterministic plan is used, its safety stock should depend on the planning day.** The decision-focused correction learned to keep a high service quantile only for the day that is actually executed and to plan later days near the 0.76–0.79 quantile. A planner can implement this insight without machine learning by tuning day-specific quantile levels; it removed about half of the waste of the deterministic quantile plan in the base case.

4. **Look at store-level service, not only at averages.** Integrated plans concentrated stock-outs at stores that are expensive to reach, whereas the inventory rules served all stores alike. Retailers with per-store service targets should constrain or monitor this effect.

5. **Decision-focused learning is a refinement, not a substitute for good inventory structure.** DFL never made the deterministic lookahead significantly worse, improved it in nine of ten settings (significantly in five) and was the cheapest integrated policy in five,

lowering stock where the quantile plan over-stocked and raising it where it under-stocked. It did not, however, beat the best inventory rule in any setting. Its value lies in correcting a fast deterministic planner at no extra online cost, not in replacing stochastic modelling or well-tuned replenishment rules.

7.2. Limitations and further research

The demand data are generated by a documented contextual model rather than observed at a retailer, so cost levels should not be read as estimates for a particular chain; testing the policies on store-level sales of fresh products, including censored demand, is the most important next step. Exact routing by enumeration limits the network to about a dozen stores; larger networks require heuristic routing, which would weaken the clean separation between replenishment and routing quality used here. Lead time is zero and each store sells one product; positive lead times would increase the value of stock-age information (Broekmeulen and van Donselaar, 2009), and multiple products would couple stores through shared vehicle space. The lookahead policies relax routing integrality beyond the implemented day, which favours visiting stores more often than a fully integer lookahead would. LA-DFL was trained on the decision states visited by LA-Quantile in one round; iterating between training and data collection, learning visit decisions as well, or training the scenario generator of ScenPO-SS decision-focusedly are natural extensions. Finally, deep reinforcement learning has shown promise for single-location perishable inventory (De Moor et al., 2022; Temizöz et al., 2025); extending it to the routing-coupled action space studied here is an open problem.

7.3. Conclusion

Replenishing perishables on shared routes is a perishable inventory problem first and a routing problem second. Under a common closed-loop simulator with exact optimisation and ten settings, stock-age-based base-stock rules were the strongest policies unless travel was expensive, in which case scenario-based stochastic lookahead was best; decision-focused learning made a fast deterministic lookahead competitive with stochastic and robust lookahead and revealed why deterministic plans mis-stock, but it did not overturn these conclusions. The structural results, the experimental protocol and the code are available for testing further policies on the same benchmark.

Data and code availability

All code, configuration files, random seeds and result files needed to reproduce every table and figure are provided in the replication package (`pirp/`, `experiments/`, `results/`). The demand data are generated by the package; no proprietary data are used.

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Appendix A. Proofs

Proof of Proposition 1. Fix v and k and let $R^\star = \{i : z_{ivk} = 1\}$. Constraint (3) gives $\sum_R y_{Rvk} = 1$ and $\sum_{R \ni i} y_{Rvk} = z_{ivk}$. For $i \in R^\star$ the second equality equals the first, so every R with $y_{Rvk} > 0$ contains i; for $i \notin R^\star$ the sum is zero, so no such R contains i. Hence $y_{Rvk} > 0$ implies $R = R^\star$, and $y_{R^\star vk} = 1$. The routing cost is $\mathrm{TSP}(R^\star)$, vehicle loads are at most Q by (4), and each store is on at most one vehicle. Every feasible lookahead plan (stores per vehicle, loads, inventory flows) has exactly one representation of this form, which proves exactness. □

Proof of Theorem 1. (i) Setting $z = 0$, $y_{\emptyset vk} = 1$, $q = 0$, $\zeta = 0$ and serving demand from available stock in any order is feasible: day-0 stock satisfies the shelf constraint by assumption and stock only decreases afterwards. All cost terms (travel, holding of carried units, waste, lost sales) are non-negative, so the objective is bounded below by zero; with finitely many assignments and a bounded linear programme for each, an optimum exists.

(ii) Take an optimal recourse solution and view inventory as fungible flows of units per cohort. *Greedy service.* If on day k demand is lost while $\varepsilon > 0$ units are left on the shelf, sell ε of them. Those units originally had one of four fates: sold on a later day (that later

sale is now lost; the cost change is $-p_u\varepsilon + p_u\varepsilon - h\varepsilon \cdot (\text{nights saved}) \le 0$), expired or disposed (saving $p_s\varepsilon$ and holding), or carried beyond the horizon (saving holding). The cost does not increase. *FIFO.* If on day k a younger cohort r' is sold while ε units of an older cohort $r < r'$ are left, sell ε older units instead and keep ε younger ones. The kept younger units replicate the original path of the older units: they are sellable on every day the older units were (their remaining life is longer); where the older units expired at the end of day $k + r - 1$, the younger units are disposed on that day at the same cost p_s and with the same number of holding nights; disposal and carrying beyond the horizon are replicated identically. The number of units carried each night is unchanged, so the cost is unchanged. Each exchange reduces $\sum_k \sum_r r\, w_{irk}$ and the feasible set is a polytope, so a finite sequence of exchanges yields an optimal solution with FIFO issuing and greedy service.

(iii) q enters the recourse LP only through the right-hand side of $S_{imk} = q_k$, so V_{LP} is the value function of an LP in its right-hand side, which is convex and piecewise linear. The physical FIFO trajectory with $\zeta = 0$ is feasible for the recourse LP and has cost C_{FIFO}; hence $V_{\mathrm{LP}} \le C_{\mathrm{FIFO}}$. By (ii) there is an optimal solution with FIFO issuing and greedy service; if it uses no disposal it coincides with the physical trajectory and the two values are equal. □

Proof of Proposition 2. (a) By LP duality, the optimal dual of the constraint $S_{imk} = q_k$ is a subgradient of the value function with respect to q_k. Adding ε to q_k and disposing of these units immediately is feasible and costs $p_s\varepsilon$ (holding $h\varepsilon$ through S_{imk} plus $(p_s - h)\varepsilon$ through ζ), so $V_{\mathrm{LP}}(q + \varepsilon e_k) \le V_{\mathrm{LP}}(q) + p_s\varepsilon$ and $\pi_k \le p_s$. Removing $\varepsilon \le q_k$ units turns at most ε sales into lost sales and cannot increase holding or waste, so $V_{\mathrm{LP}}(q - \varepsilon e_k) \le V_{\mathrm{LP}}(q) + p_u\varepsilon$ and $\pi_k \ge -p_u$.

(b) C_{FIFO} is a composition of finitely many min, max and affine maps of q, hence piecewise linear and Lipschitz. Compare the FIFO trajectories with deliveries q and $q + \varepsilon e_k$. On day k the extra units are the freshest and are sold last, so sales of older stock are unchanged. On later days the extra units are older than subsequent deliveries and may displace them, but stock that was older than the extra units is never affected. Therefore, relative to the original trajectory, lost sales can only fall, waste and carried stock can only rise, and the total extra stock E_ℓ carried on night ℓ satisfies $0 \le E_\ell \le \varepsilon$. Each extra unit leaves the system by preventing a lost sale ($-p_u$), by expiring on some day $\tau \le H - 1$ ($+p_s$ after at most $\tau - k \le H - 1 - k$ nights), or by remaining at the end ($H - k$ nights). The cost change per unit is therefore at least $-p_u$ and at most $\max\{p_s + (H - 1 - k)h,\ (H - k)h\} = p_s + (H - 1 - k)h$ when $h \le p_s$. Dividing by ε gives (7) for both one-sided derivatives. □

Proof of Proposition 3. With the visit pattern fixed and no capacity binding, the coverage map gives $q_k = \sum_{j \in W_k} \tilde{d}_j - U_k(\tilde{d})$, where U_k is the planned usable stock that serves the window. A unit increase of $\tilde{d}_j$, $j \in W_k$, increases q_k by one unless the planned stock already covers day j,

in which case the target of day j does not affect any delivery; days outside every window affect no delivery. The chain rule then gives (8), and setting it to zero is the first-order condition of $\min_q \mathbb{E}[C_{\mathrm{FIFO}}(q; D) \mid \xi]$ over the deliveries of the fixed pattern. With $m = 1$ nothing is carried, $W_k = \{k\}$, $q_k = \tilde{d}_k$ and $C_{\mathrm{FIFO}} = \sum_k p_u(D_k - q_k)^+ + p_s(q_k - D_k)^+$, so $\mathbb{E}[\partial C/\partial q_k \mid \xi] = -p_u\mathbb{P}(D_k > q_k \mid \xi) + p_s\mathbb{P}(D_k < q_k \mid \xi)$, which vanishes when $F_{D_k|\xi}(\tilde{d}_k) = p_u/(p_u + p_s)$ (for continuous F; for discrete demand, at the smallest $\tilde{d}_k$ with $F \geq p_u/(p_u + p_s)$). □